

Profiting From Sector Dispersion

The Case for Trading Sector & Industry ETFs

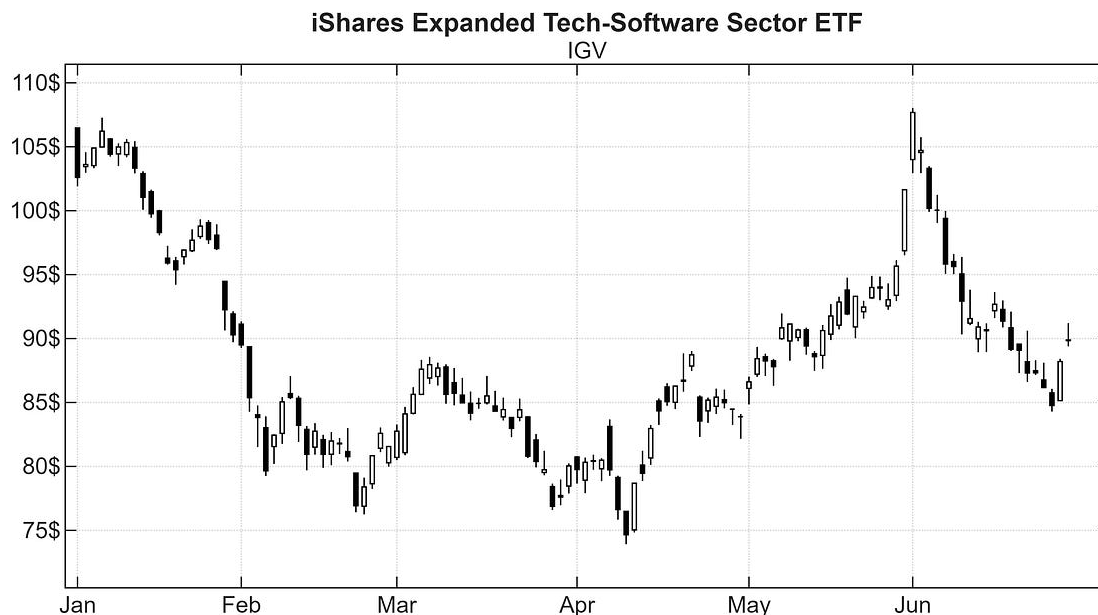


CONCRETUM RESEARCH

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These days it really seems that the commentary around **sector-rotation dynamics** has become a central topic of conversation across various social media platforms.

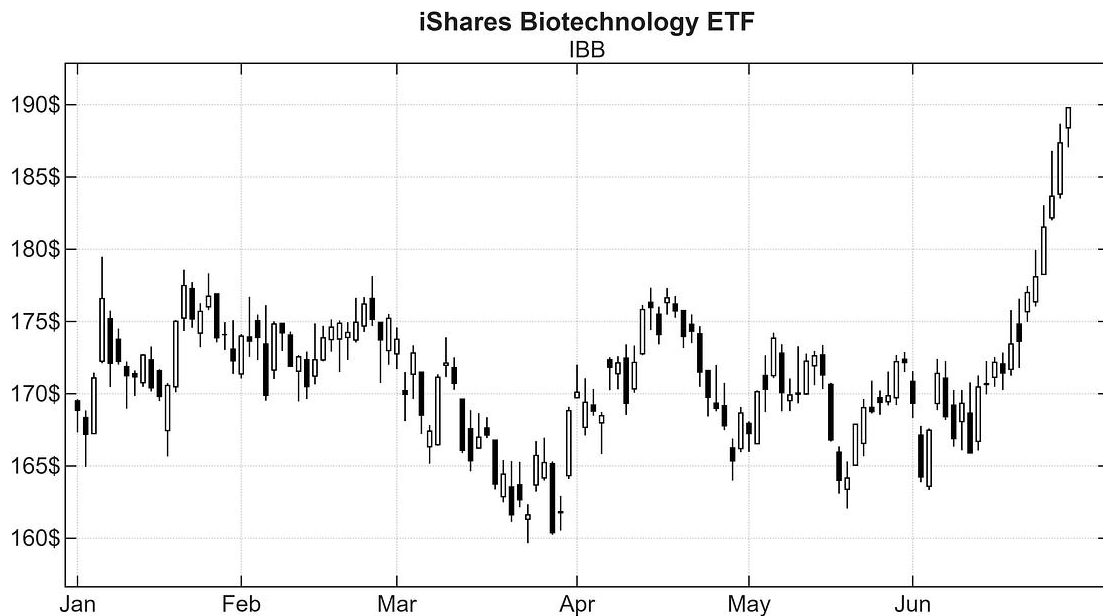
From investors debating whether the recent **drop in software-related names** makes for a compelling long-term opportunity...



...to more active market participants weighing the odds of a **correction in semiconductors** after an essentially unprecedented run...

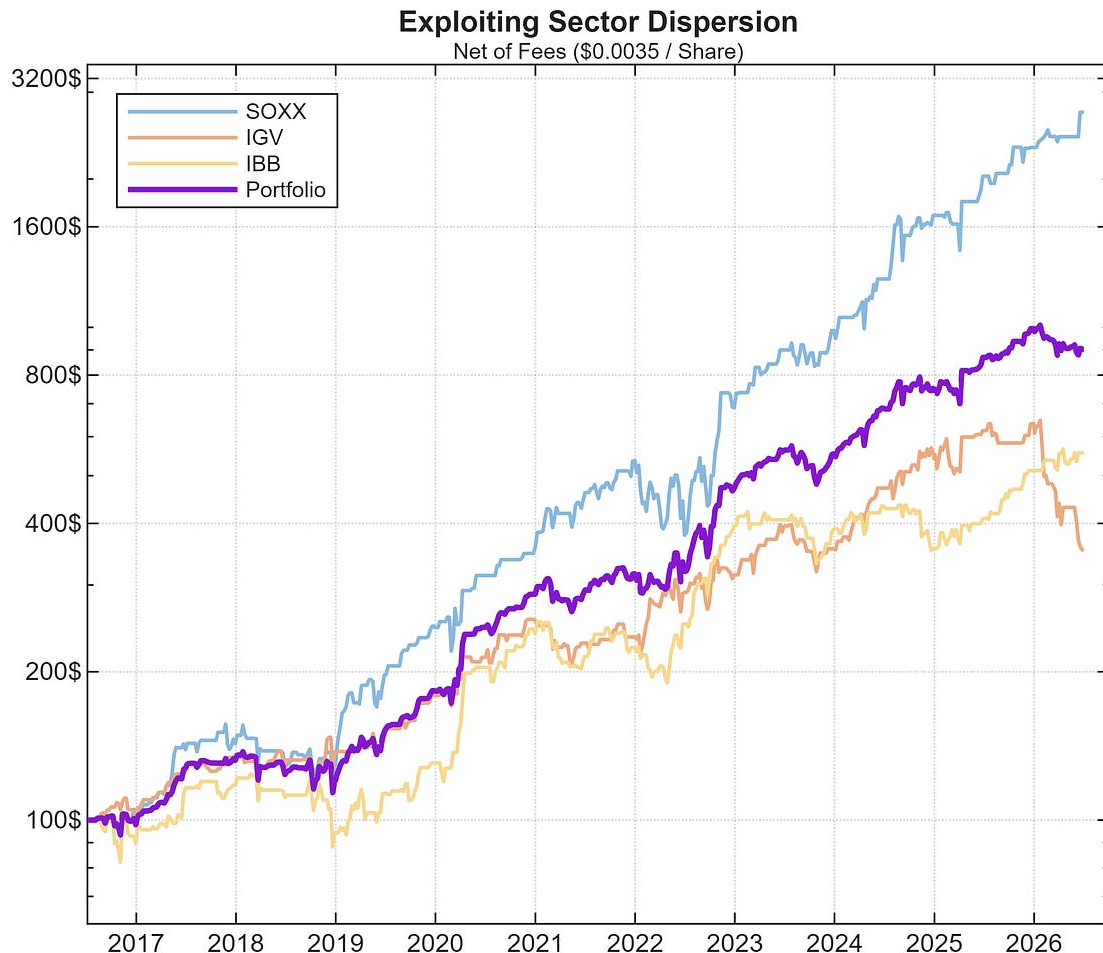


...to swing traders shifting their attention to **biotech**, which has recently set a fresh multi-year high after a long stretch of consolidation.



As a team that manages proprietary capital through quantitative frameworks, market views and commentary sit outside of what we deal with day to day, but every now and then they can **spark some interesting ideas**.

Inspired by the dispersion in sector returns we have recently witnessed, today we present a trading strategy that seeks to **exploit momentary price dislocations**, in a way that remains attractive **net of conservative transaction-cost assumptions**.



As a case study, we apply it to **three industry ETFs** that capture some of the sector-rotation “debate” mentioned earlier: the *iShares Semiconductor ETF* (SOXX), the *iShares Expanded Tech-Software ETF* (IGV), and the *iShares Biotechnology ETF* (IBB).

We show how a portfolio constructed by **actively timing** these ETFs has generated a **net-of-fee CAGR of 24.6%** and a Sharpe ratio of 1.17, while keeping maximum drawdown at ~20%.

In addition to the **rules behind the signal** and our **portfolio-construction approach**, this piece gives us a chance to touch on a delicate topic that, in our opinion, is central to the **analysis and implementation of many short-term trading strategies**.

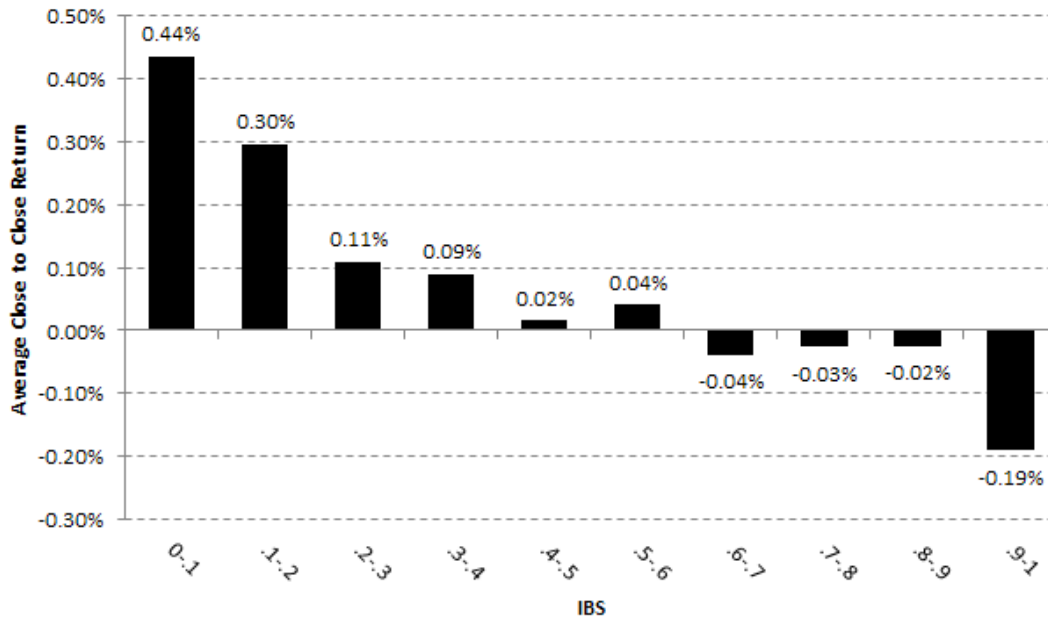
Some Literature Review

Reviewing the existing literature on mean-reversion trading, we find two contributions that point to **Internal Bar Strength** (IBS) as an effective metric for capturing this kind of price behavior in equity markets.

Specifically, IBS serves as a **proxy for directional conviction** during regular trading hours (RTH), and it is formally defined as:

$$IBS_t = \frac{Close_t - Low_t}{High_t - Low_t}$$

Pagonidis, in *The IBS Effect: Mean Reversion in Equity ETFs* (2013), documents that low IBS readings precede above-average next-day returns, while high readings precede below-average ones, an effect he shows to be present in equity ETFs since the early 1990s and remarkably consistent through time.



[*The IBS Effect: Mean Reversion in Equity ETFs \(Pagonidis, 2013\)*](#)

Additionally, Pandey and Joshi (of NYU Stern) test the idea on a basket of country ETFs in *Using Internal Bar Strength as a Key Indicator for Trading Country ETFs* (2023), confirming on more recent data that **an IBS below 0.2 tends to be followed by a bounce, and an IBS above 0.8 by a pullback.**

Signal Generation

Taken together, the evidence we have gathered suggests we should expect positive results from **incorporating IBS** information into our signal.

In particular, we define the core buy and sell conditions on symmetrical IBS readings, with entries triggered when

$$\frac{\text{Close}_{15:45} - \text{Low}_{15:45}}{\text{High}_{15:45} - \text{Low}_{15:45}} < 0.1$$

and exits triggered when

$$\frac{\text{Close}_{15:45} - \text{Low}_{15:45}}{\text{High}_{15:45} - \text{Low}_{15:45}} > 0.9$$

Since IBS is meant to capture directional conviction, we find it **economically sound** to condition our entries and exits on sessions where the price move itself was somewhat meaningful, rather than just expected “noise.”

To do so, for every ETF, we compare each day’s realized range through 15:45 with that same ETF’s own recent range history. We define the daily (log) range as

$$\text{Range}_t = \ln\left(\frac{\text{High}_t}{\text{Low}_t}\right)$$

and scale today’s value (measured through 15:45) by the volatility of the prior 21 sessions’ full-day (log) ranges, lagged by one day so no information about today’s session leaks into the scale itself:

$$\text{Relative Range}_t = \frac{\ln(\text{High}_{15:45,t} / \text{Low}_{15:45,t})}{\text{StdDev}(\text{Range}_{t-1}, \dots, \text{Range}_{t-21})}$$

Importantly, we only act on an IBS signal when this relative range value exceeds 1, i.e. only on days whose range through 15:45 already runs **above 1σ** , signaling higher-than-average range expansion.

As more experienced quants might have noticed from the formulas above, a crucial aspect of our signal generation concerns the practice of *near-close* calculation.

Computing a signal from the closing print and assuming an immediate fill at that same price technically generates no look-ahead bias, since

every input is known by the time the trade is determined, yet it introduces a hard (unrealistic) assumption: that a trade can be executed at that very price, even though no time is left in the session to actually do so. **Having seen many simulations compromised by this shortcut**, we think this is a pitfall worth calling out explicitly.

To keep the signal-generation process genuinely implementable, we therefore **extract OHLC snapshots up to 15:45** (fifteen minutes before the session closes), which leaves us enough time to compute IBS and relative range readings and submit our market-on-close (MOC) buy-sell orders.

Note - The intraday OHLC snapshot is captured 15 minutes before market close: 15:45 on normal trading days, or 12:45 on early-close days. If no price update is available at the required timestamp, the most recent prior update is used instead.

Portfolio Construction and Other Assumptions

For portfolio construction, we adopt a simple equal-weighted approach, in which the number of shares to be bought when a position is initiated is defined as

$$\text{Shares}_{i,t} = \frac{1.5 \times \text{AUM}_{t-1}}{N \times P_{i,t}^{15:45}}$$

where N represents the size of our investable universe and $P_{i,t}$ is the 15:45 snapshot price at which ETF i is purchased at the close of day t .

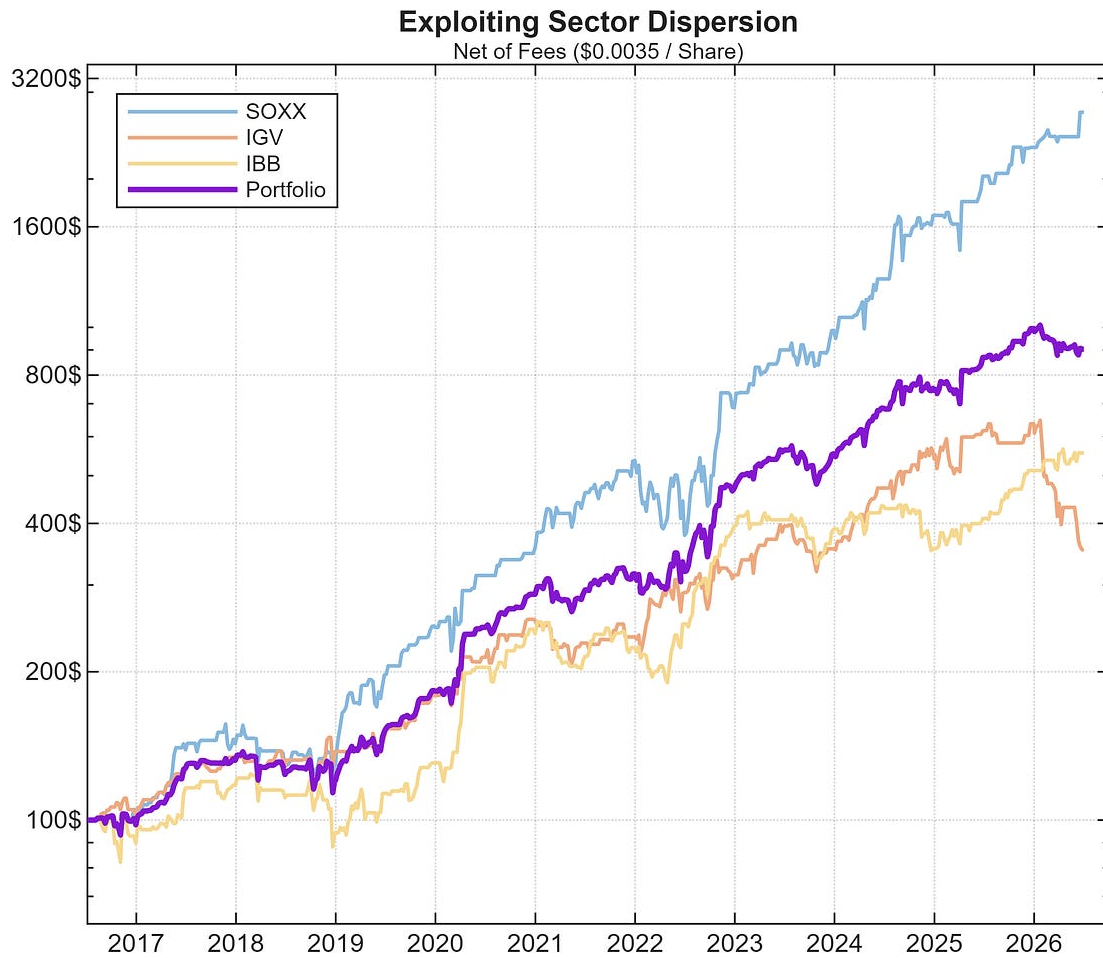
The **1.5x factor is used to gently lift the average portfolio exposure**, making up for the stretches in which the book sits partly in cash

because no entry opportunities are detected. The table below summarizes the remaining assumptions behind our historical simulation.

Assumption	Explanation
Returns	Returns include reinvestment of dividends and are adjusted for splits
Position Management	Once initiated, positions are never touched until liquidation
Fees	Standard IBKR tier of \$0.0035 / Share, doubled on sell transactions to conservatively account for additional sell-side charges (no hard minimums are applied)
Fractional Shares Trading	Trading of fractional shares is allowed
Interest on Cash	No interest earned on idle cash balance
Cost of Leverage	No financing cost is charged on leverage

Results

Over the past decade, our active timing framework has generated positive results across all three ETFs, with net-of-fee CAGRs ranging from **13.5%** to over **39%**.



Individual ETFs are sized using $N=1$, while the aggregate portfolio uses $N=3$ (see the formula presented in the previous section). Data is fetched from [Massive](#)

In our opinion, what deserves closer attention (aside from the performance of the individual sectors) is the improvement in risk-adjusted metrics **once we combine the three ETFs into a single aggregate portfolio**, rather than trading each one separately.

Strategy	CAGR	Sharpe (0% Rf)	Max Drawdown	Avg Exposure
SOXX	39.3%	1.20	-32.0%	50.8%
IGV	13.5%	0.63	-49.7%	49.2%
IBB	18.7%	0.88	-30.5%	60.9%
Portfolio	24.6%	1.17	-20.7%	53.3%

All results are presented net of fees (\$0.0035/Share, doubled for sell transactions to conservatively account for additional sell-side charges). Portfolios assume no interest earned on idle cash and no financing cost for leverage.

Notably, the portfolio's Sharpe ratio (**1.17**) sits close to that of the best-performing ETF (SOXX), while maximum drawdown (which ranges from **-32%** to **-49.7%** across the individual ETFs) falls to just **-20.7%**.

Without changing entry-exit or sizing criteria, the combined portfolio **captures dislocations that do not always occur simultaneously**: semiconductors, software, and biotech each tend to reach oversold or overbought conditions at different points in the cycle. Because such episodes unfold in a somewhat idiosyncratic (i.e. sector-specific) way, **combining them dampens portfolio-level risk without compromising returns**.

This can be observed directly in the **correlation matrix** of daily returns across the three single-ETF strategies and the aggregate portfolio.

Strategy	SOXX	IGV	IBB	Portfolio
SOXX	1.00	0.48	0.35	0.83
IGV	0.48	1.00	0.41	0.79
IBB	0.35	0.41	1.00	0.71
Portfolio	0.83	0.79	0.71	1.00

Correlation matrix of daily active returns, excluding days when both strategies have no position.

Pairwise correlations between the individual sleeves are relatively contained: **0.48** between SOXX and IGV, **0.41** between IGV and IBB, and just **0.35** between SOXX and IBB, confirming that entries and exits across the three sectors are loosely related. The aggregate portfolio naturally correlates more closely with each of its components, but its risk profile clearly **benefits from the dispersion that those modest inter-ETF correlations underscore.**

Conclusion

All in all, we believe that sector-rotation dynamics (beyond simply fueling day-to-day market commentary) are best treated as a useful characteristic of sectors and industry groups, one that can be **directly exploited when constructing actively managed portfolios.**

In particular, the appeal of industries (as opposed to individual stocks) stems from the fact that **they sit at an interesting intermediate level:** broad enough to capture dispersion in sector returns, yet specific

enough to express idiosyncratic exposure without just loading up on generic market beta.

With this in mind, we **believe sectors and industry groups can offer a compelling alternative to trading single-name equities directly**, especially given the growing number of liquid ETFs tracking them.

Whether through long-term trend-following programs, such as the one described in our 2024 paper *“A Century of Profitable Industry Trends,”* or through shorter-term strategies like the one presented here, we think genuine opportunities exist for traders willing to harness them, while **keeping operational burden and complexity at manageable levels.**



A Century of Profitable Industry Trends

CONCRETUM RESEARCH • JUNE 7, 2024

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